R15

Code No: 123BN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Obtain the PCNF and PDNF of $(P \rightarrow (Q \wedge R)) \wedge (\sim P \rightarrow (\sim Q \wedge \sim R))$. [15]

2. Show that $(S \vee R)$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$. [15]

3. Draw the Hasse diagrams of the following sets under the partial ordering relation "divides", and indicate those which are totally ordered.
   a) {1,2,3,6,12} b) {3,9,27,54} [15]

4.a) On the set Q of all rational numbers, the operation * is defined by a*b=a+b-ab. Show that under this operation, Q forms a commutative monoid.
   b) Verify the relation R= {(1,1),(2,2),(3,3),(1,3),(3,1)} is equivalence relation or not? If so find the equivalence classes. [8+7]

5.a) How many three digit numbers can be formed which are even?
   b) Find the number of binary numbers with seven 1's and five 0's. [8+7]

6.a) Determine the co-efficient of $a^2b^3c^2d^5$ in the expression of $(a + 2b - 3c + 2d)^{12}$.
   b) Find the number of integral solutions to $x_1+x_2+x_3+x_4+x_5 = 50$, where $x_1 \geq -4$, $x_2 \geq 7$. [7+8]

7. Solve the recurrence relations $a_n-2a_{n-3}+a_{n-6}=0$ for $n>=6$ and $a_0=1$, $a_1 = a_2 = a_3 = a_4 = a_5 = 0$. [15]

8.a) Define tree. Explain in detail about properties of spanning tree.
   b) Explain Kruskal's algorithm with example. [8+7]

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